KAIZEN MATHS

Lesson starter

Transposition of Formulae: Changing the Subject

Brackets and Fractions

Name _____

S1

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x + 5)$

S2

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{x+5}{6}$

S3

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{4x+2}{3}$

S4

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x - 10)$

KAIZEN MATHS

Lesson starter

Transposition of Formulae: Changing the Subject

Brackets and Fractions

Name _____

S1

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x + 5)$

S2

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{x+5}{6}$

S3

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{4x+2}{3}$

S4

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x - 10)$

ALG-0852-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Brackets and Fractions

Name _____

EA1

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{x-10}{7}$

EA2

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{5x+5}{5}$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Brackets and Fractions

Name _____

EB1

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x + 6)$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Brackets and Fractions

Name _____

EA1

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{x-10}{7}$

EA2

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{5x+5}{5}$

ALG-0852-L2 · EA

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Brackets and Fractions

Name _____

EB1

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x + 6)$

ALG-0852-L2 · EB

Show your working

Teacher answers & working

Transposition of Formulae: Changing the Subject · Brackets and Fractions

ALG-0852-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x + 5)$

Answer:

$x = \frac{y}{6} - 5$

Working:

Start with the formula. The bracket containing x is multiplied by 6.

$y = 6(x + 5)$

Divide both sides by 6 to isolate the bracket.

$\frac{y}{6} = \frac{6(x+5)}{6}$

Cancel the 6 on the right-hand side.

$\frac{y}{6} = x + 5$

Undo the addition inside the bracket.

$\frac{y}{6} - 5 = x$

Write x as the subject.

$x = \frac{y}{6} - 5$

S2

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{x+5}{6}$

Answer:

$x = 6y - 5$

Working:

Start with the formula. The whole numerator is divided by 6.

$y = \frac{x+5}{6}$

Multiply both sides by 6 to remove the denominator.

$6y = x + 5$

Now undo the addition with 5.

$6y - 5 = x$

Write x as the subject.

$x = 6y - 5$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Brackets and Fractions

ALG-0852-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{4x+2}{3}$

Answer:

$x = \frac{3y-2}{4}$

Working:

Start with the formula. The whole numerator is divided by 3.

$y = \frac{4x+2}{3}$

Multiply both sides by 3 to remove the denominator.

$3y = 4x + 2$

Subtract 2 from both sides to isolate the x term.

$3y - 2 = 4x$

Divide both sides by 4.

$\frac{3y-2}{4} = x$

Write x as the subject.

$x = \frac{3y-2}{4}$

S4

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x - 10)$

Answer:

$x = \frac{y}{6} + 10$

Working:

Start with the formula. The bracket containing x is multiplied by 6.

$y = 6(x - 10)$

Divide both sides by 6 to isolate the bracket.

$\frac{y}{6} = \frac{6(x-10)}{6}$

Cancel the 6 on the right-hand side.

$\frac{y}{6} = x - 10$

Undo the subtraction inside the bracket.

$\frac{y}{6} + 10 = x$

Write x as the subject.

$x = \frac{y}{6} + 10$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Brackets and Fractions

ALG-0852-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{x-10}{7}$

Answer:

$x = 7y + 10$

Working:

Start with the formula. The whole numerator is divided by 7.

$y = \frac{x-10}{7}$

Multiply both sides by 7 to remove the denominator.

$7y = x - 10$

Now undo the subtraction with 10.

$7y + 10 = x$

Write x as the subject.

$x = 7y + 10$

EA2

Make the named variable the subject of each formula.

Make x the subject.

$y = \frac{5x+5}{5}$

Answer:

$x = \frac{5y-5}{5}$

Working:

Start with the formula. The whole numerator is divided by 5.

$y = \frac{5x+5}{5}$

Multiply both sides by 5 to remove the denominator.

$5y = 5x + 5$

Subtract 5 from both sides to isolate the x term.

$5y - 5 = 5x$

Divide both sides by 5.

$\frac{5y-5}{5} = x$

Write x as the subject.

$x = \frac{5y-5}{5}$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Brackets and Fractions

ALG-0852-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Make the named variable the subject of each formula.

Make x the subject.

$y = 6(x + 6)$

Answer:

$x = \frac{y}{6} - 6$

Working:

Start with the formula. The bracket containing x is multiplied by 6.

$y = 6(x + 6)$

Divide both sides by 6 to isolate the bracket.

$\frac{y}{6} = \frac{6(x+6)}{6}$

Cancel the 6 on the right-hand side.

$\frac{y}{6} = x + 6$

Undo the addition inside the bracket.

$\frac{y}{6} - 6 = x$

Write x as the subject.

$x = \frac{y}{6} - 6$